

EXPERIMENT 02

Sample Java Program to Check Whether the Given Number is Prime Palindrome or Not

■ Basic Information

Experiment No: 02

Date: 11/09/2026

Title: Sample Java Program to Check Whether the Given Number is Prime Palindrome or Not

■ Objective / Aim

To develop a Java program to check whether the given number is a **Prime Palindrome** or not using Java **data types, variables, operators, control structures, methods, and console input/output**.

■ System Requirements

Hardware

- Standard PC/Laptop
- Minimum 4 GB RAM
- Minimum 500 MB free storage

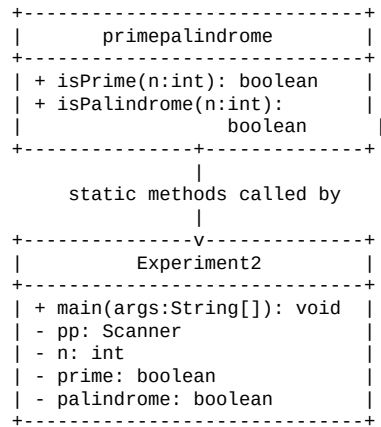
Software

- Windows / Linux / macOS
- Java Development Kit (JDK)
- Java IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
- Command Prompt / Terminal

■ Algorithm / Logic Description

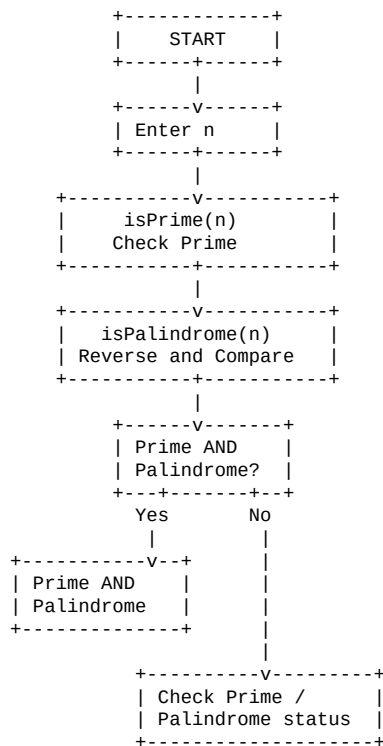
1. Start the program.
2. Import the Scanner class.
3. Create a Scanner object to read input from the user.
4. Read an integer n from the user.
5. Call the isPrime(n) method to check whether the number is prime.
6. If n is less than or equal to 1, return false.
7. If n is less than or equal to 3, return true.
8. If n is divisible by 2 or 3, return false.
9. Check possible factors using a loop starting from 5 and increasing by 6.
10. If n is divisible by i or i + 2, return false; otherwise return true.
11. Call the isPalindrome(n) method.
12. Store the original number and reverse the number digit by digit.
13. Compare the original number with the reversed number.
14. If both prime and palindrome are true, display that the number is a PRIME NUMBER AND PALINDROME.
15. If only prime is true, display that the number is prime but not palindrome.
16. If only palindrome is true, display that the number is not prime but is a palindrome.
17. Otherwise, display that the number is neither prime nor palindrome.
18. Close the Scanner.
19. Stop the program.

■ UML Class Diagram



Relationship: The Experiment2 class contains the main() method and calls the static methods isPrime() and isPalindrome() of the primepalindrome class.

■ Program Flow Diagram



■ Source Code

File name: Experiment2.java

```
package recordprog;

import java.util.Scanner;

class primepalindrome {

    // method to check prime number
    static boolean isPrime(int n) {

        if (n <= 1)
            return false;

        if (n <= 3)
            return true;

        if (n % 2 == 0 || n % 3 == 0)
            return false;

        for (int i = 5; (long) i * i <= n; i += 6) {

            if (n % i == 0 || n % (i + 2) == 0)
                return false;

        }

        return true;
    }

    // method to check palindrome number
    static boolean isPalindrome(int n) {

        int original = n;
        int reversed = 0;

        while (n != 0) {

            int digit = n % 10;
            reversed = reversed * 10 + digit;
            n /= 10;

        }

        return original == reversed;
    }
}

public class Experiment2 {

    public static void main(String[] args) {

        Scanner pp = new Scanner(System.in);

        System.out.println("Enter the number");
        int n = pp.nextInt();

        boolean prime = primepalindrome.isPrime(n);
        boolean palindrome = primepalindrome.isPalindrome(n);

        if (prime && palindrome) {

            System.out.println(
                n + " is a PRIME NUMBER AND PALINDROME"
            );

        } else if (prime) {

            System.out.println(
                n + " IT IS PRIME BUT NOT PALINDROME"
            );

        } else if (palindrome) {

            System.out.println(
                n + " IS NOT PRIME BUT IT IS A PALINDROME"
            );

        } else {

            System.out.println(
                n + " IS NEITHER PRIME NOR BE PALINDROME"
            );

        }

        pp.close();
    }
}
```


■ Compilation & Execution Commands

Since the program contains package `recordprog`;, create a folder named `recordprog` and save the file inside it as:

```
recordprog/Experiment2.java
```

Compile:

```
javac recordprog/Experiment2.java
```

Run:

```
java recordprog.Experiment2
```

■■ Sample Output

Case 1: Prime and Palindrome

```
Enter the number
131
131 is a PRIME NUMBER AND PALINDROME
```

Case 2: Prime but Not Palindrome

```
Enter the number
137
137 IT IS PRIME BUT NOT PALINDROME
```

Case 3: Not Prime but Palindrome

```
Enter the number
121
121 IS NOT PRIME BUT IT IS A PALINDROME
```

Case 4: Neither Prime nor Palindrome

```
Enter the number
123
123 IS NEITHER PRIME NOR BE PALINDROME
```

■ Result

The Java program to check whether a given number is a **Prime Palindrome** or not was successfully implemented. The program uses **methods, variables, data types, operators, loops, conditional statements, and console input/output** to determine whether the given number is prime, palindrome, both, or neither.